energyAI Berlin

Interview task: fine-tune & serve small language model

Build an end-to-end pipeline to collect domain-specific data, fine-tune a small language model, and deploy it for production use.

Input

- Target domain (e.g., "electric vehicle charging stations")
- Use case (e.g., QA)
- Data sources (e.g. PDFs)
- Base small LLM for fine-tuning (e.g. Llama-3-7B)

Output

Working AI pipeline with automated data processing, fine-tuned small LLM, evaluation and benchmarking, and production deployment and serving.

Requirements:

- Config
  - Target domain topic as prompt in NL.
  - Environment variables
- Data collection
  - Web scraping
  - PDF extraction with layout
  - Metadata extraction and source attribution
- Data processing
  - Cleaning, deduplication, and quality filtering
  - Data normalization and tokenization
  - Data storage
- Training dataset
  - LLM API integration for dataset augmentation
  - Question-answer pair generation
  - Training dataset formatting and preparation
- Fine-tuning
  - Integration with models ≤7B parameters (e.g., Llama-3-7B)
  - Memory-efficient training with techniques like LoRA or QLoRA
  - Experiment tracking
- Evals and Benchmarking
  - Domain-specific benchmark dataset creation (you should come-up with one related to the task and automatically generated)
  - Automated evaluation metrics (ROUGE, BLEU)
    1. Performance comparison with baseline model

    - Inference latency and throughput
- Deployment and Serving
    - Model registration and versioning
    - Lightweight inference and deployment
    - API endpoint with authentication
    - Endpoint monitoring
- Orchestration
    - Workflow automation
    - Manual and scheduled triggers
- MLOps - CI/CD

<u>Deliverables</u>

- Modular codebase organized by pipeline stage
- Configuration file
- Readme, how to!

<u>Criteria</u>

- Pipeline executes end-to-end without manual intervention
- Fine-tuned small LLM show improvement over baseline
- Deployment serves requests
- All pipeline components include logging and monitoring

<u>Note</u>:

You may choose tools and frameworks based on your expertise, not requirements.

You should explain your approach to prompting code generation tools for this task. E.g. How do you structure prompts? How do you handle a long context?

<u>Task submission:</u>

Send a public Github repo-link to: contact@energyai.berlin.

The task is expected to be completed until Friday 25th of July.